EUR/USD ORACLE

Future Implementation Masterplan

From the Stage 6 research foundation to ORACLE Mission Control

Edition 1.0 | 30 October 2026 | Research-first architecture | Proposed future programme

Project baseline Stage 6 engineering complete at Git commit 1408d5e on feature/utc-research-m01-m08.

Handover position 176 of 176 defined Stage 1-6 engineering checkpoints completed. The last approved Windows regression reported 1,200 passed, 14 skipped and 61 subtests passed. Eleven independent external evidence gates remain unverified, as do all thirteen source-family attestations. FINAL=HOLD.

End point of this document A working, dependable, evidence-first **ORACLE Mission Control Dashboard** that can inspect, replay and explain the research system, its Council, its sources and its risk posture. This roadmap does not authorize paper trades or capital deployment.

> ***Executive decision.*** *Our next competitive advantage will not come from adding a 32nd agent. It will come from making every prediction causally traceable, every disagreement understandable, every result independently assessable and every operational state observable. Intelligence without evidence is a hypothesis; a dashboard without trustworthy provenance is decoration. We build the evidence first, the Council research cycle second, performance proof third, adaptive intelligence fourth, and Mission Control fifth.*

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1. The destination we are building toward

ORACLE began with an ambitious but coherent idea. Instead of predicting EUR/USD using a single indicator, it would convene specialists spanning technical structure, statistical forecasting, institutional behavior, monetary policy and risk governance. The Council would not simply vote. It would reason over a common market snapshot, preserve disagreement, challenge its preferred scenario, compare alternatives, hold when evidence is weak, and learn from independently recorded outcomes.

The original Version 1.1 blueprint describes a data intelligence layer, market state engine, 31-seat Council, MiroFish scenario simulation, consensus engine, separate risk governance, execution engine and learning memory. Its intended heartbeat is a completed five-minute market cycle. The later research-first design deliberately makes financial execution a conditional downstream capability, not an assumption built into every stage.

The intended ORACLE experience, when mature

1. The system knows **what is known as of the decision cutoff**, not what was discovered afterwards.
2. It distinguishes a fresh price, stale quote, scheduled event, revised macro release, operator policy and a derived model feature.
3. Each of the 28 primary analysts submits a reproducible research opinion, while the Devil's Advocate and Risk Officer can expose weaknesses and veto action independently.
4. The Chief Strategist produces an attributable, versioned assessment rather than an opaque signal.
5. Scenario tools show plausible ranges, sensitivity and failure cases, not guaranteed price destinations.
6. A predecision prediction receipt survives unchanged until its forecast horizon matures. Later outcome evidence cannot be used to improve the earlier prediction retroactively.
7. Reliability and cost-aware comparisons determine whether Council complexity is justified. An absent advantage is reported honestly.
8. Mission Control makes all these layers inspectable through a single coherent, read-only interface, with visible HOLD states and an auditable timeline.

This is a research intelligence platform first. Paper execution, broker order routing and live-capital deployment belong to separate future authorization decisions and are outside the acceptance criteria for this roadmap.

2. Our confirmed starting point

2.1 Engineering completed through Stage 6

Engineering stage	Completed checkpoints	Major outcome
Stage 1	60	Initial infrastructure and research foundations
Stage 2	20	Forward-readiness and data lineage foundations
Stage 3	24	Research controls and

Engineering stage	Completed checkpoints	Major outcome
		acceptance contracts
Stage 4	24	Broker-clock, capture custody and Council provenance inspection
Stage 5	24	Source causality, recovery, prospective evidence and handover
Stage 6	24	Governance, evidence envelopes, point-in-time Council context, prospective ledger, evaluation and final Git/policy audit
Total	**176**	**Defined Stages 1-6 engineering complete**

These counts are from the established engineering work programme. Completion signifies implemented checkpoints and the defined regression allowlist, not an end-to-end production acceptance test or independent trading edge.

The Stage 6 commits, in order, are 9fc5e1d, 3ad7e04, e403da8, ee5a43d, 7a4bcfb and 1408d5e. Each batch introduced one runtime module and its corresponding test file. The final Windows regression for the Stage 6 allowlist was **1,200 passed, 14 skipped, 61 subtests passed**. The last supplied Git output confirmed a clean tree and matching local and origin-tracking HEAD at 1408d5e.

2.2 Immutable historical boundary

- Historical epoch ID: 75311e239cd5c295d3e08e03c358488fb123a46bfc6d342dd31493d95710aa33.
- Historical clock policy: docs/clock_profiles/hfm_forward_20261027.json.
- Original policy raw SHA-256: e34b7a29f78f5865f4b0fe7c4ad778e4e8b650d9943792eee019a48bdf2fe088.
- Last Stage 6 operator-reported Git HEAD: 1408d5e.
- Current independent external gate verification: **0/11**.
- Current independently verified source-family coverage: **0/13**.
- Binding operational state: **FINAL=HOLD**.

The October 27 historical clock and capture provenance must not be repaired retrospectively, silently rebased to UTC or presented as broker-attested. A future forward epoch, if justified, must have a separately reviewed scope and independent time/source evidence. The Stage 6 precommit Git inspector intentionally expects HEAD 7a4bcfb; it should not be rerun unchanged against 1408d5e. Stage 7 begins with a new six-commit postcommit audit rather than modifying that historical inspector in place.

2.3 What the repository already offers

The earlier repository review identified existing source implementations for all 31 Council seats, a Council manifest and typed context contracts, market data and forward storage modules, research Council orchestration components, learning and delayed-outcome contracts, model artifacts and multiple backtesting and safety-test components. A Monte Carlo agent exists; a complete integrated MiroFish-style scenario laboratory remains a future development objective. A dashboard directory was part of the blueprint's planned project structure, but an integrated Mission Control experience is not yet verified as complete.

The archived repository snapshot supplied to this project predates the confirmed Stage 6 Windows HEAD. It is useful for architecture orientation but should not be treated as an exact current checkout. All new Stage 7 source decisions must begin with the current Git tree on the Windows system.

2.4 The distinction that will guide every future stage

Question	Engineering evidence can answer	Independent acceptance must answer
Is the validator implemented?	Code review, unit tests, negative tests	Not applicable
Is a local file unchanged?	Hash comparison, Git blob integrity	Who supplied the original bytes and when?
Is a source timestamp plausible?	Chronology, UTC parsing, revision checks	Is its time backed by an independent reference?
Does a Council context have 13 family labels?	Static contract inventory and local coverage	Are the providers authentic, sufficiently fresh and admissible?
Did a prediction exist before its outcome?	Local claimed chronology	Is there an independent, immutable predecision receipt?
Is a hypothetical strategy profitable?	Exploratory arithmetic on supplied data	Does it survive genuinely frozen, cost-aware, independent evaluation?
Is an interface complete?	Visual tests, API contracts, snapshot replay	Does it display true provenance and fail closed under real outages?

Rule Nothing becomes VERIFIED merely because an internal test passes or a locally computed hash matches.

3. Roadmap at a glance

The proposed continuation is five engineering stages, each divided into six batches of four checkpoints. This creates **120 proposed checkpoints across 30 proposed batches** from P7-01 through P11-24. The numbers are planning targets, not evidence of work already performed. We will reassess stage

boundaries after each gated handover rather than committing to calendar dates without known external dependencies.

Stage	Programme	Primary result	Hard dependency	Dashboard value unlocked
7	Trusted evidence and source admission	Integrity-preserving, independently reviewable custody and source coverage	External broker, clock, provider and operator material	Source Trust and Evidence Vault views
8	Research-only five-minute Council cycle	Auditable causal shadow assessments and immutable forecasts	Accepted research scope and admissible sources	Council Room and Decision Timeline
9	Prospective performance proof	Matured outcomes and honest baseline comparisons	Independent forecast, outcome, cost and holdout evidence	Performance Laboratory and calibration views
10	Adaptive intelligence and scenario research	Testable Council improvements, historical memory and scenario explorer	Validated datasets and controlled evaluation	Model Registry and Scenario Lab
11	Mission Control dashboard and reliability	Read-only cockpit, replay, observability, incidents and operator handover	Stable versioned read models and security controls	**ORACLE Mission Control v1 delivered**

3.1 Dependency chain

Pinned history -> independently reviewable sources -> point-in-time Council context -> read-only research predictions -> independent matured outcomes -> calibrated, cost-aware evaluation -> bounded adaptive research -> Mission Control evidence cockpit

A dependency can be built with synthetic fixtures before the independent inputs arrive. Such work earns an engineering PASS and an external-evidence BLOCKED state. It does not bypass the gate. The dashboard should begin as an honest offline replay viewer, growing additional capabilities only when their underlying read models become trustworthy.

3.2 Five design doctrines

Causality before prediction The time at which evidence became available matters as much as its timestamped value. A revised economic figure must not appear in an earlier Council decision.

Diversity before majority Twenty agents repeating the same candle-derived signal are not twenty independent confirmations. Agreement will be paired with shared-source and correlated-error analysis.

Abstention is intelligent HOLD is a valid outcome, not an error state or missed opportunity. Missing sources, violated clocks, uncertain risk and incompatible regimes must be surfaced explicitly.

Reproducibility before adaptation A changing model or agent weight must have a version, training boundary, declared evaluation plan and reversible promotion record.

Observability before automation The operator should be able to reconstruct why ORACLE was silent, concerned, divided or confident long before order execution is considered.

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4. Stage 7 - Trusted evidence and source admission

Mission Turn ORACLE's existing research validators into a defensible evidence-review system. The essential shift is from "a source file exists" to "we can establish where it came from, when it was available, which version was used, how it was preserved and who approved its admission." This is the most important prerequisite for meaningful Council research.

Stage 7 readiness condition A research evidence packet, an operator-readable gate matrix, a verified postcommit code inventory and a prospective source-admission proposal. G01-G07 remain BLOCKED until authentic independent proof is supplied. The code programme can finish while external gate acceptance remains pending.

P7 Batch 1 - Recover the exact current truth (P7-01 to P7-04)

- **P7-01. Six-commit postcommit Git auditor.** Implement a new read-only audit of Stage 6's full six-commit first-parent ancestry, expected two-file scope, unchanged committed blobs, stage boundary, tracked cleanliness and local remote-tracking agreement. Do not edit the frozen Batch 6 precommit auditor. Distinguish local Git claims from independently queried remote custody.
- **P7-02. Capability inventory generator.** Discover the exact current checkout's Council, data, models, learning, API, runtime, tests and existing dashboard files without importing application modules. Classify code as implemented, tested, connected, operator-approved or externally proven. Avoid treating a class definition as a running service.
- **P7-03. Evidence custody register.** Define a bounded local index for original provider artifacts, observer identity, claimed receipt, hash, UTC availability and permission to review. Reject overwriting, symlinks, path traversal, ambiguous revisions and unsupported authenticity labels.
- **P7-04. Stage baseline attestation report.** Generate a deterministic research handover combining the postcommit Git inventory, historical policy SHA, operator-reported Windows regression and the 11-gate status. Report what was not independently checked.

Proof of done Four tests groups pass, the six Stage 6 Git anchors match the current Windows branch, no historical policy bytes change, and the generated report remains FINAL=HOLD. A new report digest is archived in the Atlas. This batch should be feasible without enabling a single external service.

P7 Batch 2 - Broker identity and UTC truth (P7-05 to P7-08)

- **P7-05. Provider identity envelope.** Specify evidence fields for broker name, server identifier, source channel, account-environment label, issuer, record ID and third-party confirmation path. Private credentials and personal account identifiers must never appear in audit exports.
- **P7-06. Independent time-reference verifier.** Accept only externally acquired, bounded time-reference observations with explicit uncertainty, source identity and receipt chronology. Separate device clock, broker timestamp, feed timestamp, capture arrival and decision cutoff. Do not infer a verified UTC mapping from a timezone string alone.
- **P7-07. Clock-domain cross-check and adversarial cases.** Test leap/offset disagreement, timezone ambiguity, delayed capture, stale sampling, future timestamps, monotonicity breaks and sample-window uncertainty. Propagate uncertainty into downstream family readiness rather than silently correcting historical labels.
- **P7-08. Gate G01-G02 reviewer packet.** Produce independent evidence requirements, missing-field detail, conflicting evidence quarantine and operator decision templates. A true external reviewer must make the acceptance decision, not the validator itself.

Proof of done Synthetic claims cannot set either gate to VERIFIED. When no independent record exists, the source identity and clock status remain UNKNOWN or BLOCKED. The original October 27 epoch is untouched.

P7 Batch 3 - Capture integrity and prospective epoch design (P7-09 to P7-12)

- **P7-09. Historical capture forensics.** Inventory original tick and candle files, first-owned capture records, sequence gaps, raw bytes, code-version hashes and archived report ancestry. Record conflicting timestamps instead of rewriting them.
- **P7-10. Source-chain reconciliation.** Verify the declared relationship among provider payload, captured bytes, candle construction, storage record and derived context. Flag gaps, duplicated observations, unsupported custody and postcapture mutation.
- **P7-11. New forward-epoch proposal.** Produce a future-only epoch charter describing identity, accepted sources, clock limits, stop conditions, retention, isolation, approval signatures and expected evidence packages. The charter itself must not start the collector or change original history.
- **P7-12. Offline preflight and recovery constraints.** Assess whether a separately authorized future session could fail safely on malformed source metadata, interrupted writes, missing clock proof or external storage failures. Preserve FINAL=HOLD unless a distinct operator decision explicitly authorizes a narrowly scoped research session.

Proof of done Historical records remain immutable, the prospective epoch has its own identity and approval boundary, and G03-G04 cannot be marked complete by a preflight simulation alone.

P7 Batch 4 - Trustworthy market and macro family coverage (P7-13 to P7-16)

- **P7-13. Thirteen-family evidence matrix.** Map every canonical Council input family to its contributing providers, transformations, freshness policy, owner, independent evidence requirement and source-specific exceptions. Distinguish direct feeds from derived context and operator policy.
- **P7-14. Release-vintage and scheduled-event engine.** Record event time, first release time, later revisions and first local availability. Permit a genuinely announced future schedule or future-effective policy while blocking future actual results from an earlier decision.

- **P7-15. Cross-family coherence rules.** Validate EUR/USD tick-to-M5 relation, cross-asset timing, COT release vintage, ECB and Fed publication boundaries, economic actuals versus schedules, and permissible missing values. Missing is not zero; unverified is not fresh.
- **P7-16. Source-admission review packet.** Generate family-by-family NOT_SUPPLIED, LOCAL_CANDIDATE, QUARANTINED, INDEPENDENTLY_ATTESTED and REVOKED classifications. Keep approval authority outside the same process that parses candidate data.

Proof of done All 13 families have a documented evidence contract, but verified coverage only increases when a qualified independent packet is supplied and separately approved. No automatic promotion from a matching hash.

P7 Batch 5 - Feature, model and policy ancestry (P7-17 to P7-20)

- **P7-17. Derived-feature dependency graph.** Map every Council-required derived label, including feature_vector, tick_features, market state and rates data, to immutable source versions and transformation-code hashes.
- **P7-18. Point-in-time snapshot builder specification.** Design an immutable, content-addressed Council context read model. Every field has observed_at, first_available_at, selected_vintage, source ancestry, freshness and uncertainty. The builder stays offline until research-only operation is authorized.
- **P7-19. Model and policy compatibility checks.** Validate forecast specification version, model artifact hash, feature schema, training cutoff, regime coverage, risk limits and evaluation-plan compatibility. Require an explicit reason when a seat abstains for absent or mismatched inputs.
- **P7-20. Causal replay and nonleakage suite.** Reconstruct historical snapshot candidates using only values available at the chosen decision cutoff. Inject revisions and late publications and prove that replay does not silently change an earlier forecast.

Proof of done Any feature in a prospective Council context can be traced back to an admitted source family and fixed transformation. G06 is not automatically accepted merely because a DAG is present.

P7 Batch 6 - Evidence board and Stage 7 handover (P7-21 to P7-24)

- **P7-21. Independent gate review interface.** Consolidate G01-G07 with evidence IDs, issuer, reviewer, verification date, scope and revocation semantics. Preserve failed, pending and not-supplied states.
- **P7-22. Risk and recovery-readiness review.** Test declared policy limits, data isolation, clock uncertainty, source revocation, safe stop, restart and escalation paths without sending any broker commands.
- **P7-23. Research source-readiness decision.** Issue a machine-readable verdict that separately reports code integrity, independent evidence, clock confidence, source-family coverage, model ancestry and operator approval.
- **P7-24. Stage 7 completion and Atlas update.** Record checkpoint commits, Windows allowlist, report hashes, evidence artifacts, unresolved gates, and the precise conditions under which Stage 8 can receive an authorized research context.

Stage 7 exit Engineering may be complete even if external admission is blocked. A real transition to research-only source ingestion requires both independent evidence and explicit operator permission. Anything less stays offline with FINAL=HOLD.

5. Stage 8 - A complete research-only five-minute Council cycle

Mission Connect the code ORACLE already possesses into a coherent, observable research process. For every eligible completed EUR/USD M5 bar, produce a fully versioned context, isolate failures across the 31-seat Council, preserve governance vetoes and register the resulting forecast before its outcome. At first, all development uses offline fixtures and archived local data. Prospective shadow operation is a separate approval step.

Boundary No broker order functions, execution adapters, live or simulated positions, model fitting or production settlement are called as a side effect of running the Council research pipeline.

P8 Batch 1 - Event-driven M5 research heartbeat (P8-01 to P8-04)

- **P8-01. Completed-bar identity and watermark.** Define deterministic IDs for epoch, symbol, timeframe, M5 close, version and source digest. Reject partial candles and duplicate bar delivery.
- **P8-02. Read-only bar admission queue.** Consume previously admitted, immutable event envelopes; check staleness, clock uncertainty, holiday/session context and provenance. Run an offline fixture first, not the historical collector.
- **P8-03. Idempotency, retry and dead-letter contract.** Exactly one accepted research cycle per decision key; retries preserve the same hash or fail. Missing bars and conflicting revisions quarantine rather than overwrite.
- **P8-04. Cycle state machine.** Define RECEIVED, VALIDATED, CONTEXT_READY, ANALYSIS_RUNNING, GOVERNANCE, REGISTERED, HELD, EXPIRED and FAILED, including legal transitions and terminal reasons.

Proof of done Duplicate or reordered data cannot produce a second independent prediction for the same decision key. Network, broker and order side effects remain absent.

P8 Batch 2 - Point-in-time Market State Engine (P8-05 to P8-08)

- **P8-05. Canonical M5 context assembler.** Populate only the source family fields available by the selected cutoff, with explicit NOT_SUPPLIED placeholders where appropriate.
- **P8-06. Regime and session attribution.** Version trend, volatility, session, event risk and liquidity calculations. Track why a regime classification changed and which source revisions it used.
- **P8-07. Family freshness and contradiction handling.** Define required versus optional fields per seat and reason-coded abstention on stale Fed, cross-asset, tick, calendar or policy input.
- **P8-08. Reproducible context hash.** A decision packet must replay into the same serialized data, code/feature version and content digest at the same evidence cutoff.

Proof of done Context construction refuses lookahead, does not silently impute absent macro data and does not mutate frozen source artifacts.

P8 Batch 3 - Council execution sandbox (P8-09 to P8-12)

- **P8-09. Seat dispatch contract.** Use the canonical 28 primary, two governance and one strategist structure. Enforce input requirements and deterministic seat IDs, not a generic pool of free-form votes.
- **P8-10. Time and resource budgets.** Bound per-seat runtime, memory, exception handling, cancellation and result serialization. One slow or failing specialist cannot hang the entire research cycle.
- **P8-11. Governance sequence.** Collect primary opinions, then Devil's Advocate challenge, then Risk Officer veto, then Chief Strategist decision. A risk veto forces HOLD regardless of agreement.
- **P8-12. Missing-seat and disagreement policies.** Decide whether incomplete Council participation invalidates the cycle, lowers confidence or yields HOLD. Do not treat absent votes as abstentions unless that substitution is explicit and audited.

Proof of done Negative tests cover timeouts, malformed votes, inconsistent seat identity, veto override attempts and nondeterministic serialization. Real Council dispatch remains disabled until operator-approved shadow scope.

P8 Batch 4 - Explainable decisions and immutable memory (P8-13 to P8-16)

- **P8-13. Decision explanation packet.** Record contributing seats, contrary views, cited source families, reliability versions, applicable risk rules and the chief decision rationale in structured fields.
- **P8-14. Consensus and correlation instrumentation.** Capture agreement and confidence while preserving shared-source ancestry. Report the effective diversity of opinions rather than assuming independent votes.
- **P8-15. Forecast envelope and no-edit rule.** Bind prediction ID, horizon, direction/probability, versioned model context and predecision registration requirements. Reject any edit after the receipt boundary.
- **P8-16. Read-only decision journal.** Produce append-only research event records, lifecycle state and reason-coded HOLD outcomes suitable for later outcome matching and dashboard replay.

Proof of done A past decision can be reconstructed without executing agents again or seeing its future label. Explanation claims point to the exact context version used.

P8 Batch 5 - Offline shadow-cycle rehearsal (P8-17 to P8-20)

- **P8-17. Historical research simulator.** Reproduce complete cycles using frozen inputs and a mock clock. Clearly label all outputs OFFLINE_REPLAY or SYNTHETIC.
- **P8-18. Prospective receipt handoff.** Define interfaces for an independent timestamp witness or receipt service without making that dependency an automatic trust oracle. No fabricated third-party receipt IDs.
- **P8-19. Fail-closed candidate shadow runner.** Package the M5 engine, Council sandbox and audit journal behind a separate RESEARCH_SHADOW_ONLY authorization gate, with no broker-order imports or execution controls.
- **P8-20. End-to-end failure rehearsal.** Simulate delayed M5 arrivals, stale quotes, late macro releases, missing seats, inconsistent epoch IDs and Risk Officer vetoes. Assert HOLD, idempotency and full evidence records.

Proof of done A deterministic offline fixture runs through the entire research path. An actual prospective cycle is not enabled without separate source and operator acceptance.

P8 Batch 6 - Shadow readiness and recovery handover (P8-21 to P8-24)

- **P8-21. Persisted cycle recovery design.** Specify exactly-once logical decision identities, replay-safe journal offsets, recovery after crashes and how partial runs are marked without duplicate forecasts.
- **P8-22. Structured observability.** Define metrics for bar lateness, seat timeout, missing families, disagreement, risk veto, prediction receipt lag and cycle completion. Do not use these metrics as proof of trading quality.
- **P8-23. Dry-run release gate.** Require approved research scope, admitted source families, an immutable forecast witness plan and a tested STOP/HOLD mechanism before any real shadow process can start.
- **P8-24. Stage 8 evidence handover.** Capture contract versions, sample decision packets, blocked transitions, test artifacts, recovery proof and the exact approved or unapproved research operating mode.

Stage 8 exit A reliable, reproducible research-only Council pipeline. Without external approval, it is an offline simulation of that pipeline, not an operational live Council. There must be no paper or live broker-order capability along this path.

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6. Stage 9 - Independent prospective evaluation and evidence of edge

Mission Determine whether ORACLE predicts usefully, when it should abstain, and whether its added complexity improves outcomes beyond transparent baselines after plausible trading costs. This stage does not assume that a strong edge exists. Its purpose is to find out, including the possibility that the correct result is "not demonstrated."

A successful engineering module can compute statistics using synthetic data. A successful research claim requires genuinely prospective custody, independently available outcomes, a frozen evaluation plan and explicit limits on what those observations can establish. Precommitted measures and negative-result reporting are first-class features.

P9 Batch 1 - Predeclared experiments (P9-01 to P9-04)

- **P9-01. Research hypothesis registry.** Predeclare target horizons, decision eligibility, forecast outputs, directional versus probabilistic metrics, abstention treatment and permissible exclusion rules before observing evaluation outcomes.
- **P9-02. Benchmark policy lock.** Include NO_TRADE, ALWAYS_LONG, ALWAYS_SHORT, momentum sign and simple source-compatible alternatives. Record why each is an appropriate comparator and which data it receives at the same cutoff.
- **P9-03. Holdout and leakage embargo contract.** Separate training, validation, calibration, challenger selection and final untouched test windows. Declare purging or embargo gaps for overlapping forecast horizons and release lags.

- **P9-04. Cost and decision policy lock.** Specify assumed bid/ask, spread, fees, slippage, financing, latency and no-trade opportunity treatment. Define the statistical and practical improvement needed before calling a new model useful.

Proof of done A candidate protocol is immutable, timestamped before evaluation and externally verifiable where claimed. Changing the protocol creates a new experiment rather than quietly updating old results.

P9 Batch 2 - Prospective prediction custody (P9-05 to P9-08)

- **P9-05. Independent forecast receipt interface.** Design a one-way interface that accepts a committed forecast hash and returns an independently verifiable registration receipt. Separate claimed local timestamps from actual external witness records.
- **P9-06. Forecast ledger integrity.** Enforce prediction IDs, decision windows, immutable signed payload, versioned model/feature ancestry and exact post-registration freeze. Reject backdating and late arrivals that claim to be predecision.
- **P9-07. Coverage and selection-bias tracking.** Register eligible M5 decisions even when ORACLE holds, fails or abstains. This prevents score inflation from reporting only winning or completed predictions.
- **P9-08. Witness-reconciliation review.** Compare independent receipt digests and original candidate payloads, detect omissions, duplications and missing witnesses, and maintain explicit QUARANTINED status.

Proof of done No internal report can promote itself to independently registered. The evaluation denominator includes every eligible decision according to the preregistered policy.

P9 Batch 3 - Outcome maturation (P9-09 to P9-12)

- **P9-09. Horizon maturation engine.** Link each registered prediction to the outcome window established before the decision. A late outcome does not modify the prior forecast.
- **P9-10. Independent quote and economic-vintage receipts.** Accept observations only with traceable provider origin, first availability, quote side, timestamp and time-domain uncertainty.
- **P9-11. Censoring and stale-outcome rules.** Distinguish missing outcomes, delayed prices, invalid market windows and legitimately unevaluable samples. Never substitute a later favorable quote after discovering the intended one was missing.
- **P9-12. Evaluation-dataset reconciliation.** Reproduce the exact set of eligible, matured, excluded and missing records, with exclusion reasons and hashes. Require that the same forecast/outcome pair feeds the Council and all baselines.

Proof of done Every analyzed outcome can be traced to an independently supplied post-horizon source. Missing outcomes remain missing, not negative returns or zero movement.

P9 Batch 4 - Costs, baselines and directional value (P9-13 to P9-16)

- **P9-13. Price-side and execution-cost model.** Distinguish price prediction accuracy from realizable bid/ask returns. Paper-assumed costs are labeled assumptions; actual broker costs require real broker-origin evidence.

- **P9-14. Forecast-vs-baseline evaluation.** Report accuracy, directional calibration, probabilistic log loss/Brier score, event-conditioned performance, coverage and abstention for ORACLE and preregistered comparators.
- **P9-15. Risk-adjusted scenarios.** Evaluate adverse moves, maximum drawdown in hypothetical portfolios, clustered loss periods, risk exposure and worst-case cost sensitivity without presenting backtest results as trade records.
- **P9-16. No-trade counterfactual.** Quantify when HOLD avoided poor conditions and when it suppressed otherwise favorable opportunities, using identical eligibility and cost assumptions.

Proof of done Any P&L display is explicitly HYPOTHETICAL unless an actual independently reconciled trade ledger supports it. A model wins only under predeclared comparisons.

P9 Batch 5 - Calibration, uncertainty and robustness (P9-17 to P9-20)

- **P9-17. Reliability and probability calibration.** Evaluate confidence bins, Brier score, expected calibration error with sample counts, calibration drift and coverage across regimes. Avoid presenting a five-sample bin as conclusive.
- **P9-18. Agent ablation and disagreement value.** Test whether removing correlated or weak seats changes out-of-sample quality. Measure whether challenge and risk veto reduce adverse outcomes under fair comparison.
- **P9-19. Dependence-aware uncertainty.** Use time-blocked uncertainty estimates and sensitivity across non-overlapping horizons; explicitly disclose small effective sample size, dependence, repeated tests and model selection.
- **P9-20. Regime, news and data-vintage stress.** Examine London and New York sessions, quiet Asian periods, high-volatility releases, rates divergences, liquidity gaps, revisions and large spreads using only the data actually available at the event cutoff.

Proof of done Each chart has a sample count, evaluation period, evidence-classification label, uncertainty note and cost assumption. No selective presentation of favorable subgroups.

P9 Batch 6 - Independent challenge and performance decision (P9-21 to P9-24)

- **P9-21. Blind challenger comparison.** Evaluate the frozen Council against preregistered simple alternatives without changing its settings after seeing the heldout test period.
- **P9-22. Multiple-testing and selection audit.** Record the number of configurations tried, abandoned baselines, lookback changes and evaluation restarts. Report how this affects confidence in any claimed discovery.
- **P9-23. Independent research review.** Provide source receipts, experiment protocol, missingness, evidence hashes, risk veto outcomes, cost sensitivity and reproducible metrics to reviewers separate from model development.
- **P9-24. Stage 9 decision memorandum.** Classify the research outcome as NOT_ENOUGH_EVIDENCE, NO_DEMONSTRATED_EDGE, CONDITIONAL_CANDIDATE or INDEPENDENTLY_SUPPORTED_FOR_RESEARCH_ONLY, with exact criteria and no automatic trading rights.

Stage 9 exit A meaningful prospective scorecard and a defensible statement about what the data does or does not show. If performance is insufficient, the appropriate work may be simplification, better evidence or stopping further complexity rather than moving automatically to Stage 10.

7. Stage 10 - Adaptive Council, historical memory and scenario intelligence

Mission Make ORACLE better for reasons that can be measured and reversed. Preserve independent research boundaries while investigating when certain agents work, which are redundant, how similar historical regimes actually resolved and what adverse future scenarios matter. The aim is not an ever-expanding ensemble. It is a smaller or smarter Council that demonstrably generalizes.

Stage 10 is conditional on having meaningful evaluation data. Algorithm development can use frozen historical fixtures, but model selection and promotion cannot cite a synthetic test as prospective improvement.

P10 Batch 1 - Evidence-grounded agent reliability (P10-01 to P10-04)

- **P10-01. Seat-level outcome attribution.** Associate each seat's predecision opinion with the matured outcome, applicable source family, regime, missingness and shared-input ancestry.
- **P10-02. Reliability windows.** Construct bounded rolling and decayed performance summaries with explicit observation counts, uncertainty and dependence controls. Prevent one lucky streak from dominating weights.
- **P10-03. Conditional competence map.** Identify when an agent has evidence of competence by session, volatility, macro environment or forecast horizon, while controlling subgroup selection and leakage.
- **P10-04. Reliability abstention rules.** Define minimum data, calibration and source compatibility required before assigning nondefault influence. Low-evidence seats remain neutral or abstain.

Proof of done Performance histories are versioned and independent of the current outcome. Every reliability weight can be traced to an eligible past sample.

P10 Batch 2 - Diversity-aware consensus (P10-05 to P10-08)

- **P10-05. Shared-ancestry graph.** Detect Council seats using the same derived features, upstream provider observations, model family or closely related methodologies.
- **P10-06. Correlation-aware aggregation.** Compare equal weighting, reliability weighting and diversity-adjusted weighting against frozen baselines using only available predecision history.
- **P10-07. Contrarian challenge analysis.** Measure whether Devil's Advocate objections and governance vetoes reveal reproducible failure modes. Do not assign veto override power to a statistical score.
- **P10-08. Consensus confidence calibration.** Turn agreement into a probability only through evaluated calibration. Track effective independent evidence count, uncertainty and HOLD coverage.

Proof of done Novel weighting must improve unseen data and survive sensitivity checks. If it does not, keep the simpler method.

P10 Batch 3 - Causal historical pattern memory (P10-09 to P10-12)

- **P10-09. As-of feature embeddings.** Define transparent pattern representations from predecision trend, structure, volatility, session, spread, event risk and macro vintage, not future outcomes.
- **P10-10. Nearest-neighbor historical search.** Search only eligible history ending before the decision, with horizon embargoes, source compatibility and duplicate-event protection.
- **P10-11. Similarity with outcome uncertainty.** Display matched case counts, regime differences, missingness, distribution of outcomes and confidence intervals. A historical analogy must never be presented as a deterministic forecast.
- **P10-12. Pattern-memory ablation.** Compare the Council with and without historical retrieval on frozen, unseen windows. Exclude retrieval if it worsens calibration or adds unstable lookalike narratives.

Proof of done The Pattern Recognition seat can cite historical examples by immutable ID, and every example respects the original decision's availability cutoff.

P10 Batch 4 - Scenario engine and MiroFish-style laboratory (P10-13 to P10-16)

- **P10-13. Scenario contract.** Define starting market state, forecast horizon, conditional assumptions, event calendar, data provenance, parameter uncertainty and deterministic random seed.
- **P10-14. Baseline scenario generator.** Extend Monte Carlo with appropriate return distributions, regime conditioning and realistic gap/volatility handling. Compare it to simple transparent alternatives before adding more complexity.
- **P10-15. Shock and counterfactual library.** Model spread widening, policy surprises, large volatility breaks, liquidity disappearance, correlated assets moving differently and failing source availability. Mark all generated paths as simulated.
- **P10-16. MiroFish integration decision.** Treat MiroFish as an optional, versioned scenario adapter or research design pattern pending technical validation, reproducibility, cost, security and benchmark assessment. Never assume a named simulation engine guarantees better EUR/USD forecasts.

Proof of done Every scenario can be replayed with the same inputs and seed. The Scenario Lab explains assumptions and failure cases, not a single magic target price.

P10 Batch 5 - Reproducible challenger models (P10-17 to P10-20)

- **P10-17. Model registry.** Bind artifact digest, training code, training data window, feature schema, hyperparameters, purpose, evaluation horizon and approver to each candidate version.
- **P10-18. Controlled offline training sandbox.** Train challenger models only on authorized, time-appropriate research data, in isolated jobs. No production model or Council weights change as a training side effect.
- **P10-19. Shadow comparison and drift monitoring.** Track challengers against incumbents on independent holdouts and prospective records, with feature drift, residual drift, calibration shifts and input-schema incompatibility.
- **P10-20. Reversible promotion and rollback design.** Define signed review packets, minimum evidence, a fixed rollback artifact and the rule that research promotion does not grant an execution permission.

Proof of done Training, assessment and promotion are separate operations. A failed challenger remains registered as a negative result.

P10 Batch 6 - Complexity and governance audit (P10-21 to P10-24)

- **P10-21. End-to-end ablation matrix.** Compare full Council, small Council, simple baseline, consensus without scenarios, historical-memory ablation and individual-model substitutions under the same sample definitions.
- **P10-22. Worst-case and regime-shift validation.** Examine parameter sensitivity, data gaps, major releases, volatility shifts and liquidity constraints, with explicitly bounded uncertainty.
- **P10-23. Complexity budget.** Account for latency, compute, memory, human oversight, source requirements and failure surface. Prefer removing redundant features to rewarding architectural complexity.
- **P10-24. Stage 10 intelligence handover.** Publish accepted methods, rejected experiments, scenario reproducibility, model registry and any independently supported performance claims. Specify the read models Mission Control will display.

Stage 10 exit An evidence-based, reversible adaptive research layer and scenario laboratory. The full Council need not outperform a simpler ensemble. If the evidence favors simplification, the project should follow the evidence.

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8. Stage 11 - ORACLE Mission Control Dashboard and operational reliability

Mission Give the operator one coherent place to understand what ORACLE knows, what it does not know, what its 31 agents believe, why the Chief Strategist holds or prefers a direction, how past forecasts matured, which evidence is independently authenticated and whether the entire research platform is behaving safely. Build a beautiful, accessible system, but design accuracy and provenance into each interaction from the beginning.

Stage 11 delivery scope An operator-facing **read-only Mission Control v1**, with immutable report views, causal replay, system-health monitoring, agent inspection, research scorecards, scenario exploration, controlled access and a reproducible deployment package. No start/stop trading button, broker login, order ticket, capital allocation control or real-time execution permission is included in the v1 dashboard.

Refinement of our earlier roadmap The earlier outline grouped dashboard work with operational reliability and future gated paper trading in Stage 11. This masterplan makes Mission Control the explicit Stage 11 acceptance milestone. Durable paper-order operations and later controlled broker deployment remain separate, conditional post-dashboard programmes that require fresh approvals. This preserves our research-first intent and prevents the dashboard project from silently becoming execution work.

P11 Batch 1 - Dashboard data contract and trust boundary (P11-01 to P11-04)

- **P11-01. Read-model inventory.** Define stable, versioned DTOs for epoch, decision, source family, market snapshot, seat vote, governance veto, forecast, matured outcome, model, evaluation protocol, gate and incident. Every record carries an as-of time and explicit evidence classification.

- **P11-02. Offline snapshot adapter.** Build a pure read-only adapter over immutable JSON/report exports and deterministic synthetic fixtures. It must never import modules that open MT5, create a database connection or dispatch Council seats.
- **P11-03. Authorization and redaction model.** Separate Viewer, Researcher, Auditor and Operator roles for future use. For v1, views remain read-only; sensitive account identifiers, credentials and unneeded private source material are redacted at export.
- **P11-04. UI evidence-safety tests.** Check unknown versus zero, revoked trust, source staleness, conflicting clocks, future timestamps, empty states, missing reports and malicious payloads. Every visual must display its underlying evidence type.

Proof of done A versioned, side-effect-free dashboard read-model API that works with zero independently verified sources, with a visible HOLD decision and no fabricated values.

P11 Batch 2 - Product shell and Executive Cockpit (P11-05 to P11-08)

- **P11-05. Visual design system.** Establish typography, spacing, status semantics, responsive layout, keyboard navigation, chart labeling and consistent attribution. Restrict color use so amber means caution, red means blocked and green appears only for grounded verified conditions.
- **P11-06. Mission Control navigation shell.** Deliver a persistent sidebar, UTC/as-of selector, epoch label, provenance mode selector, last-report digest, safe-mode banner and user-visible data freshness indicator.
- **P11-07. Executive Cockpit page.** Show M5 research cycle health, 13-family coverage, eleven-gate verification, current Chief Strategist research position, risk veto, evidence completeness, outstanding incidents and evaluation availability. Unknown statistics must be shown as NOT AVAILABLE.
- **P11-08. Reason-coded HOLD explanation.** Build a priority-ordered "Why HOLD?" panel with exact missing evidence, stale sources, governance vetoes and blocked operating permissions, each linked to the relevant inspection report.

Proof of done Every large dashboard number opens an evidence drilldown; no decorative status implies source admission. The first demo intentionally displays FINAL=HOLD and 0/11 with offline fixtures.

P11 Batch 3 - Source Trust Center and Council Room (P11-09 to P11-12)

- **P11-09. Thirteen-family Source Trust Center.** Create a matrix for canonical family, provider, first availability, freshness, last accepted report, revision status, clock uncertainty and independent reviewer state. Support source lineage expansion.
- **P11-10. Eleven-gate Evidence Board.** Show G01-G11 as independently reviewed states, not booleans inferred from local hashes. Drill down to issuer, claimed receipt, independent verification, operator approvals, conflicts and revocations.
- **P11-11. Council Room.** Display the 28 primary specialists in their technical, quantitative, institutional and macro cohorts, plus Devil's Advocate, Risk Officer and Chief Strategist. Each seat shows vote, confidence, input completeness, supporting ancestry and abstention/error reason.
- **P11-12. Dissent and influence view.** Add agent agreement and disagreement maps, shared-source clusters, diversity-adjusted consensus diagnostics and a governance veto trace. Correlated agents must not be displayed as independent confirmations.

Proof of done A user can open any Council decision, see its exact 31-seat status and trace one seat's vote back through a derived feature to a timestamped source-family candidate.

P11 Batch 4 - Decision Replay and Research Laboratory (P11-13 to P11-16)

- **P11-13. Decision Timeline.** Create a searchable read-only journal of decision IDs, as-of UTC time, input digest, forecast horizon, received/held/expired state, blocker summary, Council participation and outcome maturity.
- **P11-14. Time-Travel Replay.** Restore the same historical snapshot a seat had at its original decision cutoff. Offer an explicit split view for later revisions, without allowing those revisions to rewrite the original context.
- **P11-15. Forecast and Outcome Laboratory.** Show prediction preregistration, external receipt status, horizon maturity, excluded samples, quote provenance, cost classification and scoreable versus unscoreable cases.
- **P11-16. Evidence-first performance charts.** Present preregistered baselines, calibration plots, regime coverage, risk-veto effects, distribution of hypothetical returns and honest uncertainty. Missing evidence leaves an empty chart with an explanation, never made-up bars.

Proof of done The exact same decision can be replayed offline with matching hash and visible timestamps. The dashboard cannot present a local simulated P&L as a verified broker result.

P11 Batch 5 - Scenario Studio, Model Observatory and Incident Room (P11-17 to P11-20)

- **P11-17. Scenario Studio.** Visualize probabilistic ranges, competing future scenarios, event shocks, stress assumptions and deterministic replay seeds. Every path is visibly tagged simulated and tied to its starting snapshot.
- **P11-18. Model Observatory.** Display incumbent and challenger artifacts, training windows, feature schemas, calibration, drift, ablations and promotion/rollback records. A model marked "candidate" cannot be mistaken for an active approved version.
- **P11-19. Risk Radar and Incident Room.** Surface Risk Officer vetoes, stale data, clock disagreements, source revocations, failed research cycles, timeouts and delayed receipts with severity, first seen, last seen and human acknowledgment. Acknowledgment does not fix the cause or authorize resumption.
- **P11-20. Replay and observability comparison.** Let a researcher compare two immutable report snapshots, model versions or decision contexts. Highlight what changed and whether the difference is an actual source revision, a model update or a changed test configuration.

Proof of done All scenario charts, model comparisons and incident histories are reproducible from their records. Incidents can be reviewed and exported but not suppressed by hiding source data from the interface.

P11 Batch 6 - Secure handover and Mission Control v1 acceptance (P11-21 to P11-24)

- **P11-21. UX, accessibility and security verification.** Test keyboard navigation, screen-reader labels, color-independent statuses, smaller screens, chart fallback tables, timezone labeling, sanitized content, authorization boundaries and export redaction.
- **P11-22. Resilience and performance.** Stress-test large read-only audit archives, missing files, expired snapshots, corrupted JSON, concurrent viewers and slow history queries. The UI must render a coherent blocked state rather than hang or show stale success.
- **P11-23. Deployment and rollback package.** Define reproducible local installation, pinned dependencies, versioned read-model schema, explicit offline start command, port and authentication configuration, audit logging and a documented rollback to the last compatible dashboard build.

- **P11-24. Mission Control v1 handover.** Deliver the complete read-only dashboard, screenshots or UI tests, a screen-by-screen operator guide, proof-of-evidence checklist, test report, compatibility manifest, Atlas update and known limitations. No execution scope is created by completion.

Stage 11 exit A functioning, tested, read-only Mission Control dashboard that presents ORACLE's source trust, Council decisions, causal research history, hypothetical and independently verified results, risk vetoes, scenarios, models and operational health without crossing its no-order boundary.

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9. Mission Control product specification

9.1 Design ambition

Mission Control should feel like an engineering observatory rather than a retail trading terminal. The visual emphasis is on **truth and diagnostic clarity**, not seductive green BUY indicators. It should be useful to three audiences: the operator who needs to know whether the system is safe, the researcher who must explain predictions and outcomes, and the auditor who needs unbroken evidence ancestry.

The first screen must answer the following in less than one inspection cycle.

- What is the overall research/operational state?
- Are the input sources trustworthy, timely and causally valid?
- Has an eligible M5 decision been received, processed, held or rejected?
- What did the Council conclude, and what did its dissenters say?
- Did the Risk Officer veto the assessment?
- Is the forecast preregistered, matured and independently evaluable?
- Which evidence or approval is missing before the next mode becomes possible?

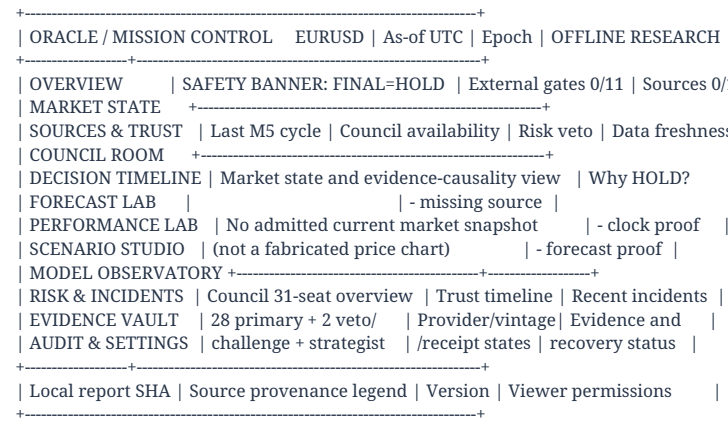
9.2 Navigation structure

Area	Primary operator question	Source of truth	Display if unavailable
01. Overview	Is the system safe and what is the latest valid research state?	Latest immutable readiness and cycle reports	HOLD plus precise missing evidence
02. Market State	What did ORACLE know about EUR/USD at this cutoff?	Versioned M5 context and source-causal snapshots	No admissible snapshot
03. Sources and Trust	Which of 13 families are verified, stale or quarantined?	Independently reviewed family evidence matrix	0/13 verified
04. Council Room	What did each of 31 seats say and why?	Frozen seat opinion ledger and context provenance	Not dispatched or unavailable
05. Decision	How did the research	Append-only cycle	No eligible cycles

Area	Primary operator question	Source of truth	Display if unavailable
Timeline	decision develop?	state and forecast records	
06. Forecast Lab	What predictions matured, and which were admitted?	Predecision receipts, quote vintages and outcome ledger	No scoreable independent samples
07. Performance Lab	Does ORACLE beat simple baselines after costs?	Frozen evaluation plan and validated research scores	No verified performance
08. Scenario Studio	What future risks and alternative paths were simulated?	Versioned scenario sets and seed/assumption pack	No scenarios generated
09. Model Observatory	Which models are candidates, approved or drifting?	Model registry and compatibility reports	No admitted model version
10. Risk and Incidents	What is blocked and what needs review?	Risk policies, vetoes and incident timeline	HOLD with unreviewed causes
11. Evidence Vault	What is the chain from a claim to its original bytes?	Artifact index, file digest, issuer and reviewer packet	Claim not authenticated
12. Audit and Settings	Which report version and operating mode am I viewing?	Immutable audit manifest and read-only user preferences	Restricted functionality

Navigation should be consistent across all pages. Every page retains the UTC as-of timestamp, historical or prospective epoch, dataset mode and evidence-verification legend. The system must never blend a replay dataset with live/prospective measurements without an unmistakable transition and labeling.

9.3 First screen layout, desktop concept



For a later independently admitted research cycle, the market-state panel can show valid, source-proven EUR/USD candles and contextual overlays. In the initial v1 offline demo it should correctly show no

admitted live chart, and optionally display synthetic fixture charts behind a persistent SYNTHETIC DATA watermark.

9.4 Interaction patterns that make the dashboard distinctive

A. Proof Breadcrumbs

Every prominent number is clickable. Selecting a Council confidence value opens a compact chain such as Decision -> Chief Strategist -> Vote set -> Seat evidence -> Context -> Family -> Provider record -> Receipt -> Hash and reviewer. This turns a polished card into an auditable assertion. An incomplete link visibly breaks the chain and marks the value unverified.

B. Why HOLD? diagnostic explainer

Rather than a generic red warning, Mission Control should rank the exact blockers affecting the selected decision. A structured explanation might say that the Fed release vintage is not independently authenticated, the broker clock offset is uncertain and the Risk Officer requires HOLD. Each reason links to the relevant gate, source envelope or policy. The interface should also explain the conditions necessary to clear a blocker without offering an unsafe bypass.

C. Decision Time Machine

A researcher chooses a historical M5 cutoff and sees exactly what ORACLE knew then. A second pane can show what was learned later, including revised economic actuals and outcome receipts. The UI must visibly distinguish **known at decision** from **learned afterwards**. This is both an analytical advantage and a guardrail against hindsight bias.

D. Dissent Map

The Council should be visualized as groups and dependency clusters, not a wall of 31 identical avatars. A matrix shows which seats voted up, down or abstained, how reliable each had been under prior data, which agents share inputs and what the Devil's Advocate challenged. It exposes false confidence caused by many agents reacting to one common feature.

E. Evidence Thermometer

Trust is never a single unverifiable percentage. Instead, show independently attributable layers such as clock, provider origin, local hash integrity, point-in-time vintage, reviewer state and authorization. A file may pass local integrity but fail provider origin. The graphic must preserve those differences.

F. Scenario Lens

A scenario is a bundle of declared assumptions, sampled paths and shock definitions. Users can compare a normal-volatility scenario to a wider-spread or unexpected ECB/Fed outcome, while the interface explains which assumptions changed. No scenario may be represented as observed price history or as a guarantee.

G. Evidence-aware Empty States

No data is a valid finding. Instead of zero-valued return charts, Mission Control explains NOT_SUPPLIED, INSUFFICIENT_SAMPLE, NOT_MATURED, QUARANTINED,

NOT_AUTHENTICATED or NOT_AUTHORIZED. This avoids the common mistake of making a sophisticated dashboard look more complete than the research evidence allows.

9.5 UI status model

Use separate, nonoverloaded fields for observation and approval.

Field	Allowed semantics	What it must not imply
Evidence status	NOT_SUPPLIED, LOCAL_CANDIDATE, HASH_MATCHED, QUARANTINED, INDEPENDENTLY_ATTESTED, REVOKED	A local hash never equals provider authentication
Data freshness	CURRENT_FOR_DECLARED_SCOPE, STALE, UNKNOWN, FUTURE_DATED, GAP_DETECTED	A recent timestamp does not establish authentic origin
Cycle status	NOT_STARTED, RECEIVED, VALIDATED, HELD, FAILED, EXPIRED, RESEARCH_REGISTERED	Registered does not mean executed or profitable
Forecast maturity	NOT_REGISTERED, PENDING_HORIZON, OUTCOME_MISSING, MATURED_UNVERIFIED, EVALUABLE	Matured without receipt is not independently evaluable
Governance	NO_ASSESSMENT, CHALLENGED, VETOED, HOLD, RESEARCH_DIRECTION_ONLY	A research direction is not an order
Operating mode	OFFLINE_REPLAY, AUTHORIZED_RESEARCH_SHADOW, PAPER_SIMULATION, LIVE_EXECUTION	A displayed mode label does not grant its permissions

For this roadmap the delivered v1 operating mode is OFFLINE_REPLAY, with AUTHORIZED_RESEARCH_SHADOW views available only if separately admitted source and shadow evidence already exists. PAPER_SIMULATION and LIVE_EXECUTION are visible as disabled future concepts, not selectable operating functions.

9.6 Visual identity proposal

An understated dark research cockpit works well because data hierarchy matters more than decoration. Suggested nonbinding design tokens are deep midnight navy for the canvas, slate panels, off-white main text and cool blue accents for navigation. Reserve amber for review required, red for BLOCKED/HOLD when safety is the issue, and green for independently grounded verified states. Avoid green for mere

code test success beside an unverified market source. Every status also needs a word and an icon, not color alone.

Typography should be clean and compact. The first screen should have strong typographic hierarchy, generous empty space and restrained charts. Use a readable monospace face for hashes and timestamps. Long receipt IDs should copy correctly and truncate visually without losing access to the complete value.

9.7 Dashboard architecture and technology choice

Proposed v1 stack A local-first Python dashboard with a pure-Python read-model layer and a Python-native UI framework, with Streamlit as an initial candidate subject to dependency and security review. This keeps early development aligned with our two-Python-file copy-and-paste workflow. It is a technology proposal, not a claim that Streamlit is already in the repository or has been selected.

Alternative for a later multiuser deployment A separately authenticated backend read API and a dedicated frontend may be appropriate if concurrency, granular permissions, richer replay interactions or remote access requirements exceed the local dashboard. Such a redesign should occur only after a documented build-versus-maintain decision and should not silently break the two-file batch rule.

Read boundary

Immutable approved research snapshots -> explicit export/redaction adapter -> versioned read models -> dashboard cache -> display components

The dashboard must not connect directly to an execution client, import an MT5 provider, create a trading session or mutate learning/forecast tables. If a secure live research source is eventually approved, its data should be exposed through read-only, bounded, versioned projections with explicit freshness and authentication labels.

Caching Cache keys must include epoch, decision cutoff, artifact hash, verification state and parser version. A changed source revision, revoked attestation or changed experiment specification invalidates the associated display cache. Silent reuse of stale "green" status is forbidden.

Security Use local-only binding by default; validate imports, paths and payload sizes; scrub secrets from logs, exports and traces; implement role-aware read access before remote exposure; add an explicit audit log for report views and exports. Do not rely on visual disabled buttons as the only security boundary.

Compatibility Keep record schemas versioned. New fields must not silently reinterpret prior snapshots. Any data contract evolution should include migration/replay fixtures and a backwards-compatibility matrix.

9.8 Suggested read-model fields

A minimal, versioned DecisionView would carry decision_id, epoch_id, symbol, m5_close_at_utc, as_of_utc, council_manifest_sha256, context_snapshot_sha256, evidence_mode, cycle_status, final_research_action, risk_veto, seat_opinions, missing_source_families, blocked_gates, forecast_receipt_id, forecast_maturity_status, evaluation_status and report_sha256.

Each SourceFamilyView should include family, is_derived, provider_reference, observed_at_utc, first_available_at_utc, selected_vintage, local_content_sha256, independent_attestation_reference, verification_status, freshness_status, clock_uncertainty and reviewer_decision.

Each SeatView should include seat_number, agent_id, stage, required_families, missing_inputs, opinion_direction, confidence_if_defined, model_version, shared_ancestry_cluster, abstention_reason, failure_reason and vote_evidence_digest.

A chart must receive typed values and an explicit evidence classification. If its chart input is unavailable, it renders an empty-state explanation. It must never substitute zeros, last-known values or fabricated demo content in an actual-evidence view.

9.9 Mission Control acceptance tests

A v1 dashboard passes only when each of these can be demonstrated using bounded fixtures and, where authorized, genuinely independently sourced report exports.

1. Opening the dashboard with no external evidence displays 0/11, 0/13 and HOLD without errors or invented candles.
2. A local hash-matched but untrusted source is visibly distinct from an independently attested source.
3. A predecision context hides a macro release that was published after the decision, even if the release event time is earlier.
4. The Council Room always reconciles 28 primary seats, two governance seats and the Chief Strategist, with missing seats clearly identified.
5. A Risk Officer veto cannot be hidden by filtering or overwritten by a positive strategist score.
6. The Decision Time Machine restores the exact original snapshot hash and does not inject later data into the earlier view.
7. A synthetic five-trade profit chart carries a permanent synthetic/hypothetical label and cannot appear in the verified-performance view.
8. Corrupted report JSON, missing artifacts, a revoked source and an unexpected schema version all fail visibly and safely.
9. All timestamps have explicit UTC labels; no timezone conversion changes decision causality.
10. Access and exports reveal no credential values or sensitive private account identifiers.
11. Every prominent KPI has a reproducible provenance path to a record or a reason why proof is missing.
12. The UI cannot start a collector, dispatch the Council, settle outcomes, train a model or submit an order.

10. Delivery method, acceptance and implementation control

10.1 Preserve our established coding rhythm

The engineering workflow remains **four checkpoints per batch, one new runtime Python file and one paired test Python file**, with the full source code provided directly in our conversation for copy-and-paste into VS Code. No ZIP-based code handover is needed. Each batch follows this sequence.

1. Confirm current Git branch, HEAD, clean tracked state and relevant source contracts.
2. Identify four explicit checkpoint outcomes and their scope boundary.
3. Present the full runtime module and full paired test file directly in chat.
4. Compile and run focused synthetic/negative tests in the Windows (oracle_env) environment.
5. Run the defined cumulative regression allowlist, which must be revised explicitly as the project grows.
6. Run a real read-only inspection with existing candidate evidence or an honest NOT_SUPPLIED result.
7. Review test output, report SHA, blockers and whether the code preserved FINAL=HOLD.
8. Stage exactly the two intended files, check the staged diff, commit and push, then confirm clean local/remote-tracking heads.
9. Update the Implementation Atlas and this Masterplan's progress record after **every four-checkpoint batch**.

The dashboard stage may ultimately require framework assets, templates or static CSS in addition to Python. Before the first such batch, deliberately decide whether to keep Python-only UI modules or authorize a narrow exception with an explicit file manifest. Do not quietly violate the batch scope for convenience. UI tests must be genuinely useful, not just assertions that a component returns a string.

10.2 Definition of done by evidence level

Level	Meaning	Example	Status in current project
E0	Design only	Proposed dashboard wireframe	Future
E1	Implemented in source	Python adapter with unit tests	Common in Stages 1-6
E2	Engineering validated	Windows focused and defined regression tests	Stages 1-6 programme complete
E3	Local custody verified	Immutable candidate hash, source AST, Git ancestry	Partially established
E4	Externally attested	Independent origin, time and reviewer packet	Not established for the eleven gates

Level	Meaning	Example	Status in current project
E5	Prospectively evaluated	Independent forecast receipts and matured outcomes under frozen protocol	Not established
E6	Operationally approved	Separate risk, source, mode and operator authorization	Not authorized

These levels are cumulative in purpose but not interchangeable. An E2 dashboard visualization must not display E4 or E5 claims unless corresponding evidence is present. A locally tested shadow runner is not an E6 operating permission.

10.3 Quality gates for every batch

Code quality Input validation, explicit types at boundaries, bounded resource use, testable pure functions and deterministic serialization where appropriate.

Causal correctness UTC event times, first available times, vintage selection, approved epoch identity and source ancestry are carried through every data transformation.

Safety The module cannot authorize collection, Council operation, settlement, training, paper orders or live orders by interpreting a local file as approval.

Adversarial tests Duplicate IDs, malformed JSON, symlinks, path traversal, missing provider fields, clock disagreement, retroactive revisions, self-promoted approval and unknown operating modes fail closed.

Operator clarity Every assessment separates confirmed engineering status from external evidence and provides a stable machine-readable report digest.

Documentation Record purpose, input/output schema, source dependency, exact command, Windows result, skipped-test rationale, Git commit and unresolved limitations.

11. Critical dependency and risk register

Risk	Why it matters	Planned control	Workstream
Incorrect broker UTC assumption	Alters candle boundaries, events and forecast horizons	Independently authenticated clock evidence and uncertainty propagation	Stage 7
Incomplete provider origin	Corrupts market/macro interpretation and trust claims	Evidence issuer, receipt and third-party review	Stage 7

Risk	Why it matters	Planned control	Workstream
Historical repair temptation	Can turn flawed old data into falsely pristine evidence	Immutable historical epoch and prospective-only replacement policy	Stage 7
Lookahead from revised releases	Makes backtests appear better than executable decisions	Point-in-time vintage and first-availability selection	Stages 7-9
All 31 seats sharing signals	Artificially exaggerates consensus confidence	Shared-source graph, pairwise errors and ablation	Stages 8-10
Selective forecast reporting	Omits poor predictions and distorts observed accuracy	Register every eligible cycle, including HOLD and failures	Stages 8-9
Transaction-cost optimism	Turns hypothetical profitability into unrealistic expectation	Bid/ask, latency, spread and cost sensitivity with independent receipts	Stage 9
Uncontrolled adaptive tuning	Overfits model selection and invalidates holdouts	Preregistered experiments and reversible model registry	Stage 10
Scenario theater	Beautiful paths mistaken for reliable probabilities	Simulation assumptions, seeds, calibration and no guarantee labels	Stage 10
Dashboard trust theater	Green badges disguise missing independent evidence	Proof Breadcrumbs, evidence-state model and empty-state tests	Stage 11
Hidden data writes in UI	Dashboard accidentally triggers collectors or trading services	Pure read models, isolated adapters and no execution imports	Stage 11
Credential leakage	A local research UI can expose broker or personal identifiers	Redacted exports, auth boundary and security tests	Stage 11
Premature execution	Code milestones mistaken for permission to deploy	Separate operator authorization, E6 gate and invariant FINAL=HOLD	All stages

12. The acceptance decision after every stage

Stage completion should produce two distinct records.

Engineering completion record What was implemented, which code and contract hashes were tested, what Windows regression passed, which issues remain and which exact Git commit was pushed. A stage can pass here.

Research/operational acceptance record Which independent source and reviewer evidence exists, whether prospective forecasts and outcomes are authentic, whether empirical improvements survive frozen testing and what operating scope is separately authorized. A stage can remain blocked here, even when its code is excellent.

The next stage should not infer permission from the previous stage's engineering acceptance. If the independent gates remain at 0/11, the operational recommendation is still HOLD. A dashboard release does not grant research-source admission, broker connection or paper/live execution rights.

12.1 Suggested human review agenda

For each four-checkpoint handover, the operator should be able to answer these questions without reading hundreds of lines of code.

- Which four capabilities were implemented, and which old contracts did they depend on?
- What is demonstrably different in a negative test, especially when evidence is missing?
- Which provider, clock, source or model claims remain independently unverified?
- Could a new feature cause a side effect through an import, UI callback or background process?
- Can the result be replayed from a fixed report digest?
- Did we change a historical anchor, run an unapproved data collection or accidentally create executable trade intent?
- What is the exact next dependency, and who or what can provide the missing evidence?

13. Immediate next action and near-term deliverables

Start at P7-01 to P7-04

The next engineering batch should be deliberately narrow. It needs no live broker call, no historical collector restart and no new market prediction. It should contain only the following two source files when the current checkout is confirmed.

Planned file	Responsibility
backend/runtime/ stage7_postcommit_evidence_baseline.py	Six-commit postcommit audit, stage capability inventory, historical pins and evidence-custody baseline
backend/runtime/ test_stage7_postcommit_evidence_baseline.py	Synthetic Git histories, source-integrity negative tests, absence-of-evidence contract and deterministic HOLD assertions

These are **proposed filenames**, not files that have been created or committed. The precise implementation must be checked against the current Windows repository after pulling no unreviewed changes.

Expected P7-01 to P7-04 operator runbook

1. Confirm `git status --short` is empty and `git log -1 --oneline` reports `1408d5e` or a separately documented successor.
2. Inspect the six Stage 6 source/test pairs and their first-parent commit chain without relying on the old precommit checker.
3. Verify the October 27 policy raw-byte pin; stop on any mismatch.
4. Run focused Windows tests, the extended cumulative regression, then the read-only assessment.
5. Share terminal outputs before Git commit. Commit only the agreed two files after acceptance.
6. Add the actual commit, report SHA, tests and remaining blockers to the Atlas and this Masterplan's progress history.

Stage 7 to dashboard progress ledger

Stage	Planned checkpoints	Implemented	Windows verified	Committed	Independent acceptance
Stage 7	24	0	0	0	Pending external evidence
Stage 8	24	0	0	0	Not authorized for real Council cycles
Stage 9	24	0	0	0	No independently validated prospective study yet
Stage 10	24	0	0	0	No accepted adaptive challenger or MiroFish integration yet
Stage 11	24	0	0	0	Mission Control not implemented or validated
Future total	**120**	**0**	**0**	**0**	**Research and operational acceptance remains conditional**

This is the current planning baseline, not a claim that every one of the 120 proposed checks is indispensable or that stages are guaranteed to proceed regardless of evidence. The programme can be refined by documented change control as discoveries are made.

14. What happens after Mission Control

The dashboard is our explicit destination for this document, but it is not the project's ultimate theoretical endpoint. If trusted evidence, prospective research and risk review warrant it, the original blueprint still contemplates paper trading and controlled deployment. Those are **separate future programmes** and must begin with their own scope, approvals, independent execution controls, broker reconciliation design and failure-test evidence. They must not be started as a side effect of completing Stage 11.

A future paper lifecycle would need its own order-intent contract, broker-independent simulator, fill and position accounting, spread/slippage assumptions, reconciliation, risk exposure reservations, restart recovery, incident escalation and independent proof that it introduces no capital risk. Controlled live deployment would require a distinct and much higher standard of third-party operational review, security, execution-boundary risk limits and human authorization.

We should celebrate a working Mission Control as a major achievement in its own right. At that milestone, ORACLE will have moved from a set of clever models and cautious validators toward a transparent, disciplined research intelligence platform with an auditable memory of what it knew, what it predicted and what really followed.

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15. Source basis, status and future updates

Original project vision *EUR/USD ORACLE: Autonomous Multi-Agent Consensus Trading Intelligence System*, Complete Engineering Blueprint, Version 1.1, seven-page project PDF. The document establishes the nine-layer architecture, five-minute research cycle, 31 Council roles, original eight-phase roadmap, database concepts and intended dashboard directory.

Current engineering record *EUR/USD ORACLE - Stage 6 Implementation Ledger and Final Handover*, 30 October 2026. The record lists the six Stage 6 Git commits, twelve Python paths, Windows regression, immutable historical clock policy, current HOLD decision and eleven unresolved external gates. Current Windows-state claims rely on the operator's supplied PowerShell output rather than an independent current checkout or GitHub-server query.

Research-first amendment The previously agreed Version 1.2 design principle is to build and validate research intelligence, independent outcomes and risk governance before considering autonomous financial execution.

Document authority This masterplan is a new **proposed Stage 7-11 implementation design**, not a modification of the original blueprint, an operator authorization, a new epoch approval or an assurance of profitable performance. Any changes to scope must have their own checkpoint IDs, rationale, dependency impacts and committed documentation updates.

Revision convention After each completed four-checkpoint batch, append a dated progress entry containing the batch name, exact file paths, Windows tests, Git commit, deterministic report SHA, proof-of-done outcome, newly cleared or still blocked evidence gates and implications for the next batch. Preserve prior entries instead of overwriting inconvenient findings.

Final guiding rule Build what can be proved. Explain what cannot yet be proved. Preserve HOLD when the evidence is insufficient. The dashboard should make this discipline visible, not conceal it.